Example 17 A man wants to fill $3\frac{3}{4}\ kg$ of rice equally in 3 bags. How much rice does each bag contain?

Solution Each bag contains $\frac{1}{3}$ of $3\frac{3}{4}\ kg$ of rice.

That is $\frac{1}{3}\times 3\frac{3}{4}=\frac{1}{3}\times\left(3+\frac{3}{4}\right)=\left(\frac{3}{3}\right)+\left(\frac{1}{3}\times\frac{3}{4}\right)$

$= 1+\frac{1}{4}=1\frac{1}{4}\ kg$

Think

$2\frac{1}{4}\times 3$ is not equal to $6\frac{1}{4}$. Why?

Example 18 In a juice shop, if a man prepared $1\frac{1}{2}$ litres of juice from $1\,kg$ of oranges, then how many litres of juice can be prepared from $12\frac{3}{4}\ kg$ of oranges?

Solution The quantity of juice prepared from $1kg$ of oranges $= 1\frac{1}{2}$ litres

The quantity of juice prepared from $12\frac{3}{4}\ kg$ of oranges $= 12\frac{3}{4}\times 1\frac{1}{2}$

$= \frac{51}{4}\times\frac{3}{2}=\frac{153}{8}$

$= 19\frac{1}{8}$ litres

Try these

i) Simplify: $35\times\frac{3}{7}$

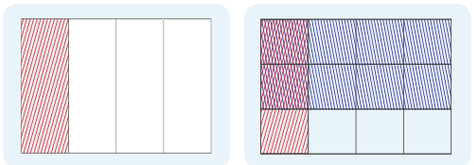
ii) Find the value of $\frac{1}{5}$ of 15.

iii) Find the value of $\frac{1}{3}$ of $\frac{3}{4}$

iv) Multiply $7\frac{3}{4}$ by $5\frac{1}{2}$

Activity

Take a paper. Fold it into 4 parts vertically of equal width. Shade one part of it with red. Then, fold it into 3 parts horizontally of equal width. Shade two parts of it with blue. Now, you count the number of shaded grids which have both the colours.



(Hint:The number of grids shaded in both colours out of the total number of grids gives the product $\frac{2}{12}$ of $\frac{2}{3}$ and $\frac{1}{4}$)

1.7 Division of Fractions

Think about the situation 1

A camp was organized in a school in which12 students participated. The camp leader wanted to divide them into groups of 2 students. How many groups were there?



There were 6 groups which was got by the division of 12 by 2.

That is 12 ÷ 2 = 6 which means there are six 2's in 12.

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